

Report for ISP Vs Real-Time Facial Recognition

CE6461 – Computer Vision Systems

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Introduction:

The experiment that is conducted within the framework of this lab is aimed at exploring the impact of camera Image Signal Processing (ISP) parameters on the quality and stability of a real-time face recognition system implemented on a Raspberry Pi platform.

The work adds to an existing recognition pipeline an explicit measure of confidence collected when comparing two facials which is based on the distance between features encoding a facial i.e. Euclidean distance, which allows readers to examine the stability of detections and accuracy of identification under imaging conditions.

Core ISP, through the Picamera2 library, is configured by programming, and then pictures are allowed to stabilize before the streaming commences, and relied upon to provide a regulated base concerning the observable influence of altered core ISP values, namely brightness and contrast, upon recognition confidence and the general behavior of the system.

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This best-match distance is then converted to a percentage confidence score which is interpretable and higher when the features are similar and smaller when the match is poor.

The experiments are conducted in the manner of one variable at a time to isolate the effect of each setting, set of default baseline against higher and lower levels whilst observing the stability of the detector, confidence variations and defeat circumstances.

Findings are only described by use of representative images and a concise discussion of relating image quality variables, like saturation and features loss at extreme settings, to variations in confidence and strength of the punch-line on the HOG-based recognition pipeline.

The derived report summarizes these findings to determine feasible operating domains of ISP control in normal contexts as well as to encourage conscience tuning to

Default Settings:

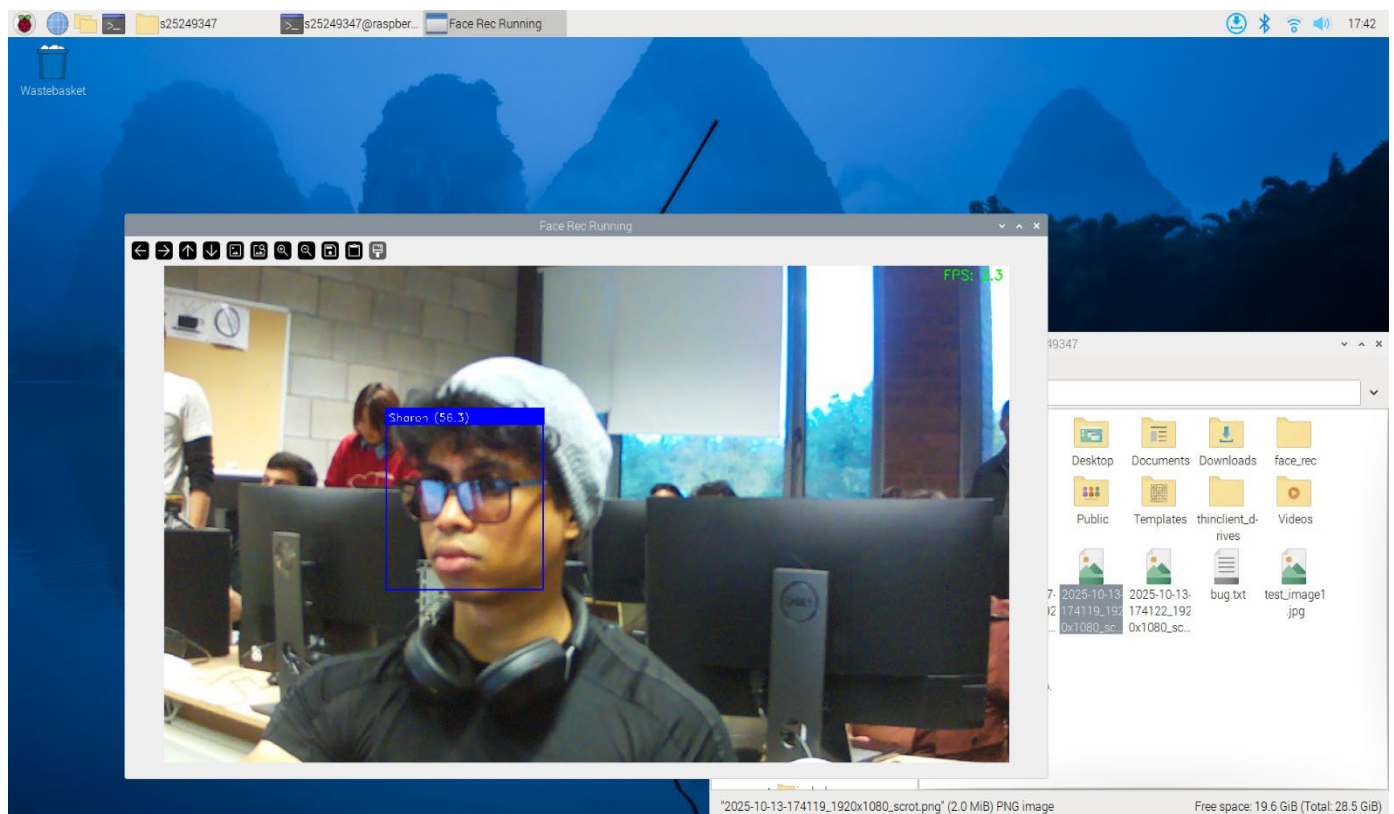


Figure 1: Defaults ISP Settings

Non-Default ISP Settings:

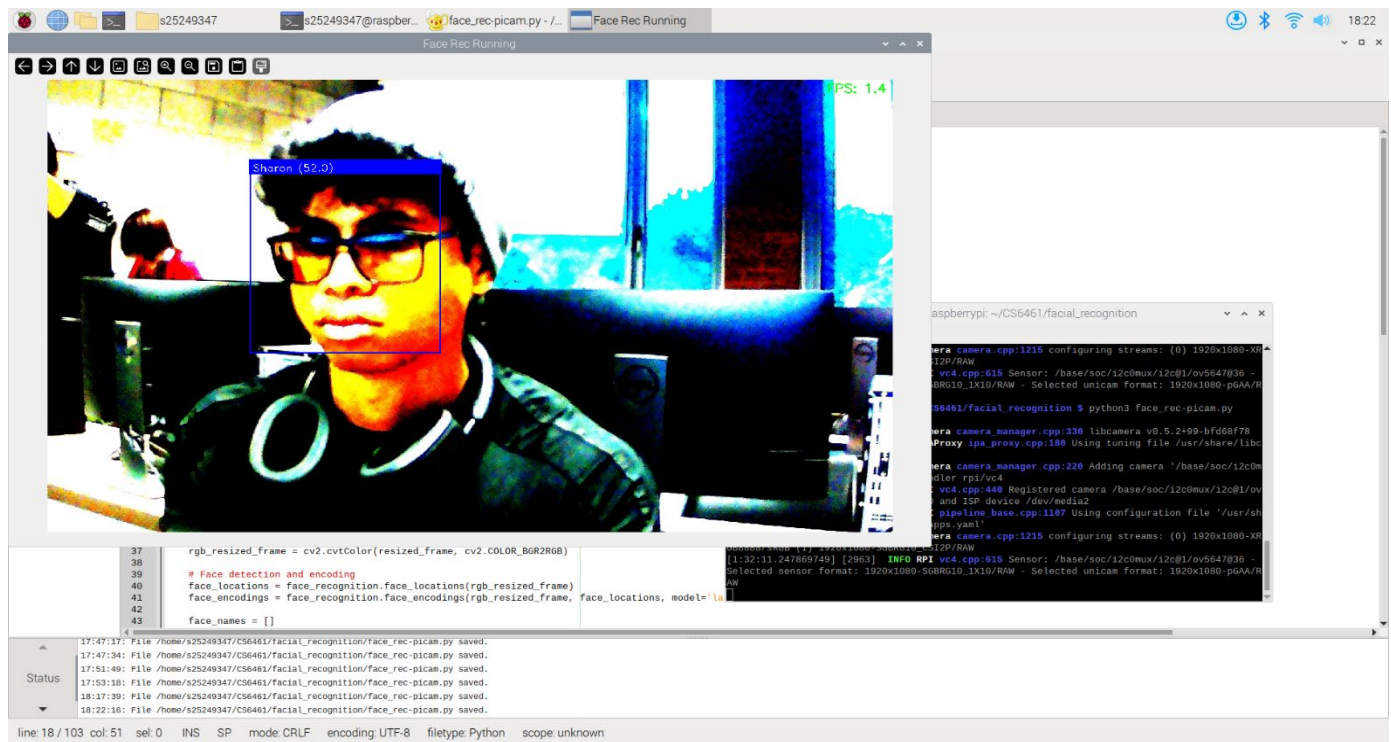


Figure 2: Brightness (0.5), Contrast (8)

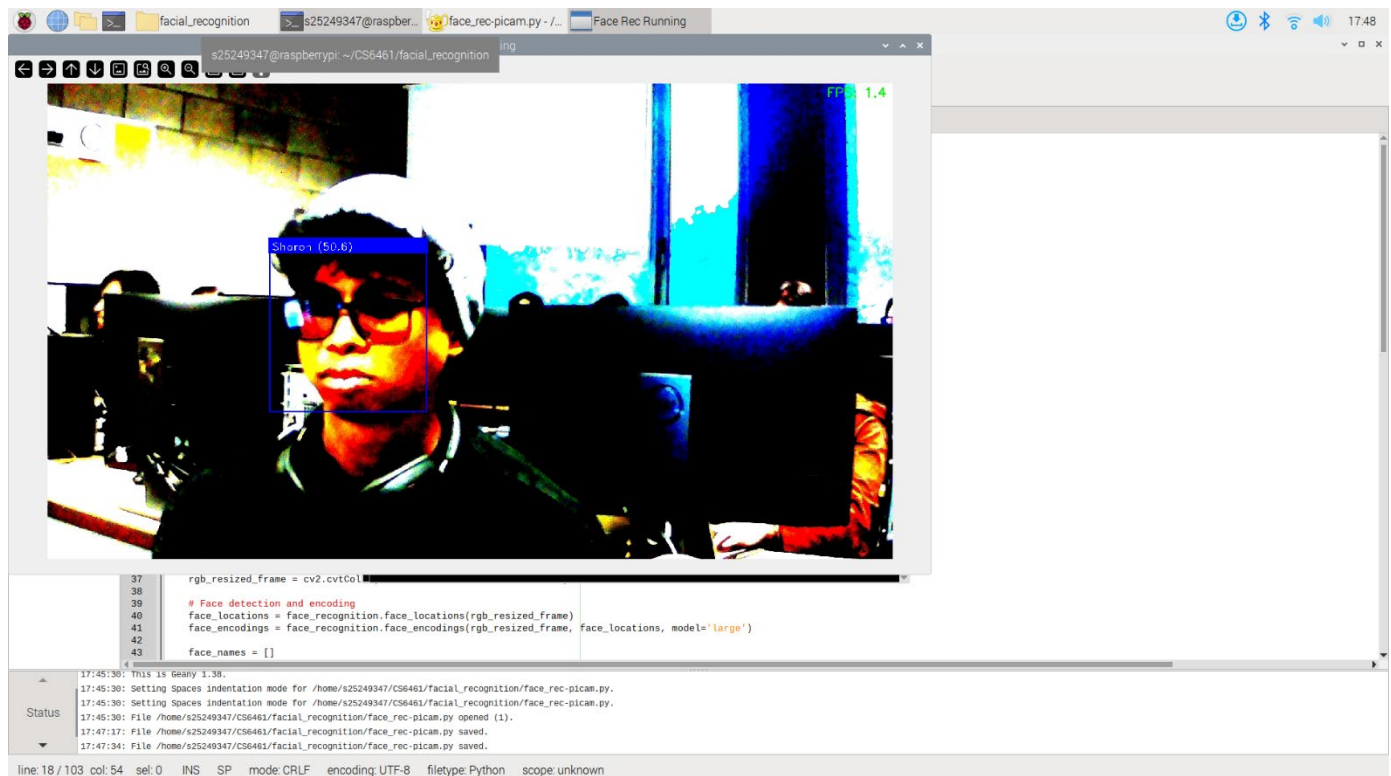


Figure 3: Brightness (1), Contrast (16)

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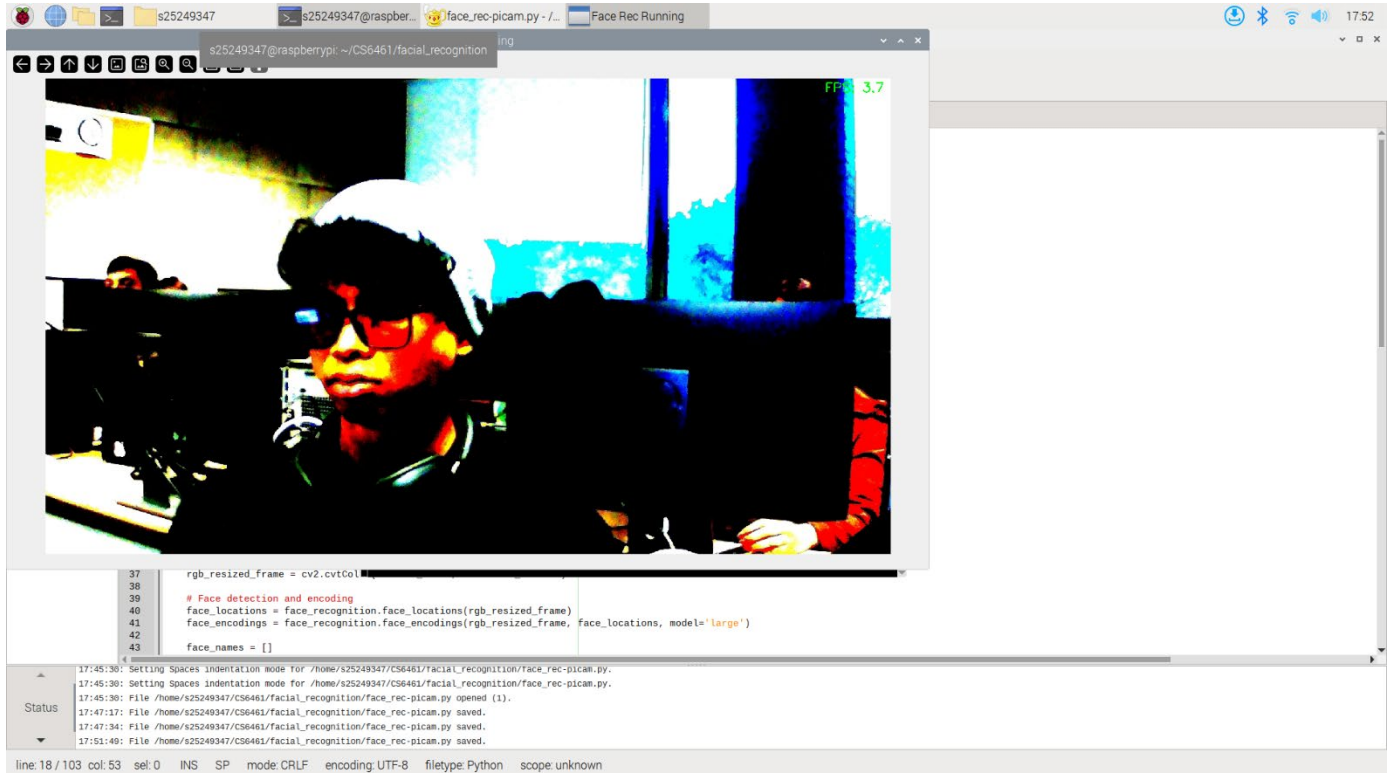


Figure 3: Brightness (-1), Contrast (32)

Experience in Script Modification:

The stages of adjusting the face recognition script to display real-time confidence scores and resizing the window was simple and without technical challenges. The necessary functionality, the visualization of the bounding box, the name of the subject, and the scores of confidence, was provided as it was stated in the instructions of the laboratory.

ISP Setting and Confidence Score Relationship:

The ISP (Image Signal Processing) parameters of Brightness (B) and Contrast (C) can be changed directly affecting the confidence score of face recognition, which is the similarity of the detected face and the stored reference image.

Optimal Range Observed:

The highest scores on face recognition with the greatest level of confidence were gained in moderate settings: B=0.5, C=8. In this case, the facial features were bright and clear and the image was not masked with over-saturation or low shadows. Limit values such as B=1, C=16 or B=-1, C=32 resulted in either over saturation or undersaturation and reduced the quality of features and reliability of the model.

Picamera 2 ISP modules In detention:

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The experiment conducted investigated the following Picamera2 runtime camera controls using Python:

Brightness: Regulates the brightness of all the pixels in the picture.

Contrast: Alters contrast between light areas and dark areas.

These parameters were adjusted through the set controls method, and configuration objects directly adjusted main and lores image streams, which have an effect on preview and video output modes.

The reason Extreme Values have an impact on recognition:

Extremely brightness results in saturation of an image, i.e. pixels are on the maximum density, and no longer reflect detailed textures on a face, resulting in loss of feature information. On the other hand, extremely low contrast causes edges and gradients to be indistinct and minimises the use of Histogram of Oriented Gradients (HOG) descriptor which computes feature vectors.

HOG calculates facial features as gradients and even directions of the cells. When ISP values dampen or saturate these gradients (e.g. flat or peaked area) these gradients reduce the discriminatory power of the descriptor, which has a detrimental impact on recognition confidence.

Additional Information

Headwear Effect:

Train images were not trained with a beanie and therefore, test images with a beanie resulted in some partial occluded train facial features, resulting in higher Euclidean feature-space distance and lower confidence scores.

Environmental Factors:

Detection accuracy was also influenced by environmental factors of light in the environment and contrast in the background since shadowing or Glare in the background further complicated HOG-based feature extraction.

Conclusion:

The experiment demonstrated that the image signal processing (ISP) settings used, i.e. brightness and contrast, have a rather significant impact on the reliability and accuracy of a face recognition system based on Raspberry Pi. We concluded that the most appropriate setting to work with was in the moderate settings of brightness and contrast, where we had the most appropriate visibility of the features and low artifact and noise. Excess or excessive contrast between these settings when taken to the extremes, lowered saturation of an image or deteriorated detail, which resulted in lower recognition performance, lower confidence scores, and instability during detection. In addition, it was observed that training and testing conditions that involved the introduction of a beanie in both

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tests and not training images resulted in the considerable reduction in recognition confidence. It highlights the need to ensure that the signifiers of training information correspond to those of the plausible real-life settings like accessories or various lighting conditions. The results reveal the importance of mindful architecture and vegetation in adoption of facial recognition applications on resource limited infrastructures like Raspberry Pi. Generally, the paper has determined that a balance between the parameters of ISP and consistency of training and application environment is needed in the quest of powerful and practical face recognition. It is possible to make future experiments and installation in the real world using such results, where flexibility and dependability are significant.

References:

- 1) Raspberry Pi Camera Team. "The Picamera2 Library", Raspberry Pi Documentation, <https://datasheets.raspberrypi.com/camera/picamera2-manual.pdf>
- 2) N. Dalal and B. Triggs, "Histograms of Oriented Gradients for Human Detection," Computer Society Conference on Computer Vision and Pattern Recognition 2005, https://en.wikipedia.org/wiki/Histogram_of_oriented_gradients
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